CS 571 Deliverable 1: Movie ROI Explorer

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Q1. Dataset Description

We use the TMDB 5000 Movie Dataset, a publicly available dataset published on Kaggle and built from The Movie Database (TMDB) API (kaggle.com/datasets/tmdb/tmdb-movie-metadata). It contains about 4,800 films, mostly U.S. productions released between 1916 and 2017.

What it represents: Each row is one movie. *tmdb_5000_movies.csv* holds title, release date, runtime, budget and worldwide revenue (USD), genres, keywords, production companies and countries, original language, popularity, average user rating, and vote count. *tmdb_5000_credits.csv* holds the full cast (with billing order) and crew (with job titles such as Director), joined to the first file by movie ID.

Type: Public and community-sourced. TMDB users contribute and edit the records, and the Kaggle release is a free snapshot of that data.

Why it matters: Making a movie is a high-risk investment. Budgets range from under $1 million to over $300 million, and many films never earn their budget back. Independent producers with limited money need evidence about which kinds of projects tend to pay off. This dataset links what a movie cost to what it earned, along with the creative choices behind it, such as genre, timing, director, and cast.

Scope for visualization: The data mixes every major attribute type: temporal (release date), categorical (genre, studio, language), quantitative (budget, revenue, rating, runtime), and relational (cast and crew links). It is large enough to show real patterns but small enough to explore interactively in a browser.

Known limitations: Over 1,000 movies list a budget or revenue of 0, which means missing, not zero. These rows are excluded from ROI views. Dollar amounts are not inflation-adjusted, and revenue is box office only.

Q2. User Identification

Our primary user is the independent film producer. Two secondary users share the same data but need different views of it.

Persona 1: Maya, independent film producer

- **Background:** Runs a small production company making one or two films a year with budgets of $2–20 million. Knows the industry well but is not a data analyst; comfortable with spreadsheets, not code.
- **Motivations:** Pick projects likely to earn back their budget, and justify those choices to investors.
- **Tasks:** Compare return on investment across genres and budget levels, check whether release timing matters, and find directors whose films are profitable at her budget scale.
- **Context of use:** At her desk while reviewing scripts or preparing an investor pitch, a few times a month.
- **Challenges:** Blockbusters dominate box office totals and hide patterns at her budget level; she has little time and no statistics training.

Persona 2: Daniel, studio acquisitions analyst

- **Background:** Acquisitions team at a mid-size distributor; strong quantitative skills; reviews dozens of finished films at festivals each year.
- **Motivations:** Decide which finished films to buy distribution rights for, and set a fair offer price.
- **Tasks:** Find past movies similar to a candidate film (genre, budget, runtime, cast) and see how much they earned.
- **Context of use:** Before and during film festivals, under tight deadlines, comparing several candidates side by side.
- **Challenges:** Must filter by many attributes at once and judge whether a few outlier hits skew the comparison.

Persona 3: Priya, film investor

- **Background:** Private investor funding a portfolio of films; financially literate, not an industry insider.
- **Motivations:** Spread risk and know how often films lose money.
- **Tasks:** See how outcomes (loss, break-even, big return) are distributed by genre and budget.
- **Context of use:** Checking a producer’s claims when a project is pitched to her.
- **Challenges:** May over-trust averages that hide high risk.

Q3. Task Identification

Figure 2 lists ten tasks (T1–T10), each in the user's own words and tied to the goal it serves. Maya's tasks are about choosing what to make and when to release it: which genres return the most per dollar (T1), which pay off at her $5–15M budget (T2), whether summer or holiday releases earn more (T3), which directors stay profitable on small budgets (T4), and whether a genre's returns are rising or falling (T5). Daniel's are about finding comparable films (T6), judging whether a few hits skew them (T7), and whether reviews predict earnings (T8). Priya's are about risk: how often a genre loses money (T9) and which genres are safe bets versus long shots (T10). The IDs are reused in Q4.

Q4. Data and Task Abstraction

Data abstraction

The dataset is a table (one item per movie) combined with a network (movies linked to the people who worked on them). Figure 1 shows the structure, and Figure 3 lists every attribute with its type: categorical (genre, company, language), ordered temporal (release date), quantitative sequential (budget, revenue, runtime, rating), and network links (cast, crew). Budget and revenue are highly skewed, and a value of 0 means missing.

Derived attributes: We compute these because the users' questions are about them, not the raw values:

- ROI = revenue ÷ budget and profit = revenue − budget: quantitative, diverging at break-even (ROI 1.0, profit $0). ROI is shown on a log scale because a few films return 50× or more.
- Outcome class (loss < 1, modest 1–3, hit > 3) and budget tier (five levels, under $5M to over $100M): ordinal.
- Release month and season (cyclic temporal) and release year (ordered temporal).
- Director track record: film count and median ROI per director, aggregated from the network.

Important structures: (1) Movie–person links form a bipartite network, which lets us aggregate results per director or actor. (2) Genre is multi-valued, so a movie is counted once in each of its genres. (3) Budget and revenue are strongly correlated, which is why we analyze ROI instead of raw revenue.

Task abstraction

Figure 4 maps each domain task to an abstract action (what the user does) and target (what it is done to), using Munzner's task typology. In short: T1, T3, and T10 compare ROI distributions across genres or seasons; T2 and T6 first filter or query by budget, genre, and runtime; T4 and T7 identify extremes and outliers; T5 and T8 discover a trend and a correlation; and T9 summarizes a distribution into outcome shares.

Across users, the core operations are filter (by genre, budget, date), compare distributions, find outliers and extremes, and look up details on individual movies. This points to a coordinated-views dashboard: shared filters drive a distribution view (ROI by genre), a temporal view (by month and year), a scatterplot (budget or rating vs. ROI), and a ranked list of directors, with details on demand for any movie.

Figures

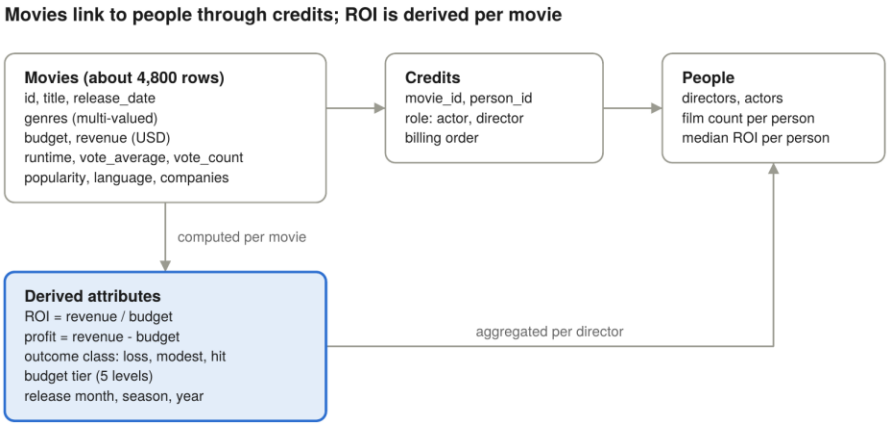


Figure 1. Data structure and derived attributes. Credits link each movie to its cast and crew. ROI, profit, outcome class, budget tier, and release season are computed per movie, then rolled up per director for task T4.

ID	User	Task (in the user's words)	Goal it serves
T1	Maya	"Which genres make the most money back for every dollar spent?"	Choose what kind of film to make
T2	Maya	"At my budget, say $5–15 million, which genres actually pay off?"	Avoid patterns driven by blockbusters
T3	Maya	"Does releasing in summer or around the holidays change how much a movie earns?"	Plan the release window
T4	Maya	"Which directors keep making profitable films on smaller budgets?"	Choose a director to approach
T5	Maya	"Is a horror film's return getting better or worse over the last 20 years?"	Check that a trend still holds today
T6	Daniel	"Show me past films like this one: same genre, similar budget and runtime."	Find comparable films for an offer
T7	Daniel	"How did those comparable films do, and are one or two hits skewing the picture?"	Set a fair, defensible price
T8	Daniel	"Do well-reviewed movies earn more, or are ratings and box office unrelated?"	Decide how much to weigh festival buzz
T9	Priya	"Out of all the dramas made, how many lost money?"	Understand the downside risk
T10	Priya	"Which genres are safe bets and which are long shots with big payoffs?"	Balance her portfolio

Figure 2. Analytical tasks for each user, in the user's own words (Q3).

Attribute	Type	Notes
Movie ID, title	Categorical (key)	Unique per movie
Genres	Categorical, multi-valued	A movie can have several (e.g., Action + Comedy); about 20 genres
Production company, country, language	Categorical	Many levels; small ones grouped into "Other"
Release date	Ordered, temporal	1916–2017
Budget, revenue (USD)	Quantitative, sequential	Highly skewed; 0 means missing
Runtime (minutes)	Quantitative, sequential	
Average rating (0–10), vote count, popularity	Quantitative, sequential	Ratings with few votes are unreliable
Cast, crew	Network links	Movie–person links with role (actor, director) and billing order

Figure 3. Data attributes and their types (Q4).

ID	Abstract task: action → target	Data used
T1	**Compare** → distribution of ROI across categories	Genre, ROI
T2	**Filter** by budget tier, then **compare** → distributions	Budget tier, genre, ROI
T3	**Compare** → distribution of ROI across a cyclic attribute	Release month/season, ROI
T4	**Locate** and **identify** → extremes among aggregated network nodes	Director, film count, median ROI
T5	**Discover** → trend over time for one category	Year, genre, ROI
T6	**Query** (search by features) → set of similar items	Genre, budget, runtime
T7	**Summarize** → distribution of a subset; **identify** → outliers	ROI of selected films
T8	**Discover** → correlation between two quantitative attributes	Rating, ROI
T9	**Summarize** → distribution; **derive** → share in each outcome class	Outcome class, genre
T10	**Compare** → spread (risk) and center (typical return) across categories	Genre, ROI

Figure 4. Domain tasks mapped to abstract visualization tasks (Q4).